SAP MM S/4HANA 2023

Open PO Monitoring & Aging Report

Portfolio Project – Training System
System: SAP S/4HANA 2023
Training System Module: SAP MM / Inventory Management
Note: This is developed in the SAP Training system for demo purposes

Designed and implemented a custom SAP MM report to monitor open Purchase Orders and track aging of outstanding PO line items. The solution was built using ABAP programming, providing real-time visibility into overdue and pending PO lines across plants and purchasing organizations.

Business Requirement

Procurement and finance teams lacked a consolidated view of open Purchase Orders and their aging status. Standard SAP transactions (ME2M, ME2N, ME2L) provided list views but required manual filtering and offered no aging classification. This made it difficult to identify overdue deliveries, follow up with vendors, and manage procurement commitments effectively.

Objectives

- Monitor all open PO line items across plants and purchasing organizations
- Classify PO lines by aging status using traffic light indicators (Green / Yellow / Red)
- Identify overdue deliveries based on scheduled delivery date vs. system date
- Support vendor follow-up and procurement escalation decisions
- Provide a reusable, executable report via custom T-code

Scope

In Scope: Open PO lines with pending delivery quantity, aging classification by delivery date, vendor details, plant and company code visibility, user execution via T-code.

Out of Scope: Automatic vendor communication, GR posting from report, integration with Ariba or external procurement systems.

Solution Overview

Custom ABAP report (ZOPEN_PO_AGING) built by joining:

- EKKO – Purchase Order header data
- EKPO – Purchase Order item data
- EKET – Delivery schedule lines (delivery dates)
- LFA1 – Vendor master

Traffic light logic applied to flag aging status per PO line. Output rendered via **REUSE_ALV_GRID_DISPLAY** with zebra striping, column optimization, and Excel export capability. Custom transaction code created in SE93 for direct end-user execution.

Functional Logic

- User enters Company Code, Plant, and optional PO Number range on the selection screen
- Report fetches open PO lines from EKKO/EKPO joined with LFA1 and EKET
- Deleted lines (LOEKZ) and fully received lines (ELIKZ) are excluded from output
- Open quantity calculated as PO Quantity minus GR Quantity; lines with open qty ≤ 0 are removed
- Aging calculated in days using system date (SY-DATUM) minus scheduled delivery date (EINDT)
- Traffic light assigned per line: Green (0–30 days), Yellow (31–90 days), Red (90+ days)
- Output displayed in ALV Grid with sort, filter, and export capability

Selection Screen Fields

Company Code, Plant, Purchase Order Number (range)

Output Fields

Status (Traffic Light), PO Number, Item, Vendor, Material, Description, PO Quantity, GR Quantity, Open Quantity, Delivery Date, Age (in Days)

Business Benefit

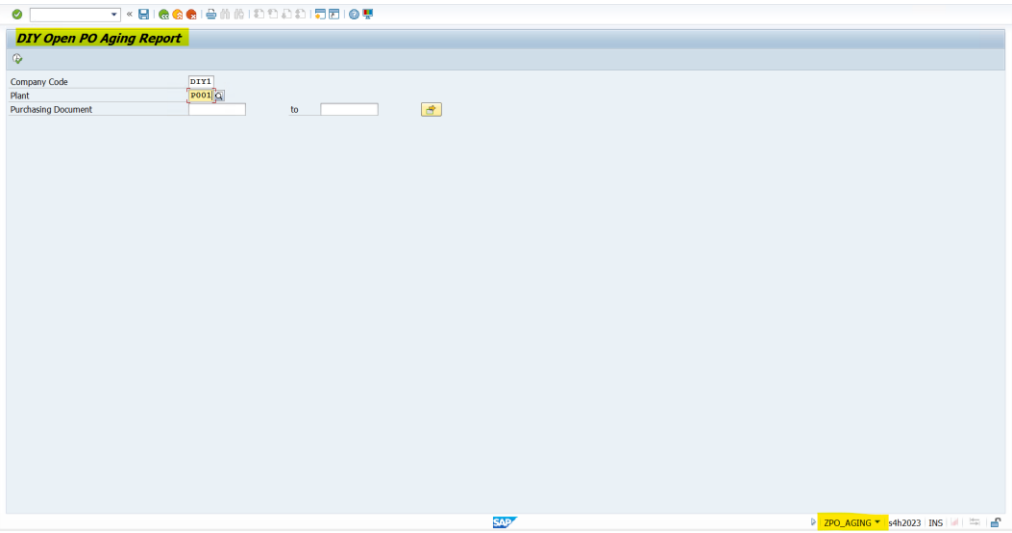
- Replaces manual cross-referencing across ME2M, ME2N, and ME2L with a single executable report
- Traffic light indicators allow procurement teams to instantly prioritise follow-up without manual date calculations
- Overdue line visibility reduces risk of missed deliveries impacting production or warehouse replenishment
- ALV Grid output supports direct Excel export for reporting and escalation workflows
- Demonstrates ABAP development capability applied to a real procurement monitoring use case

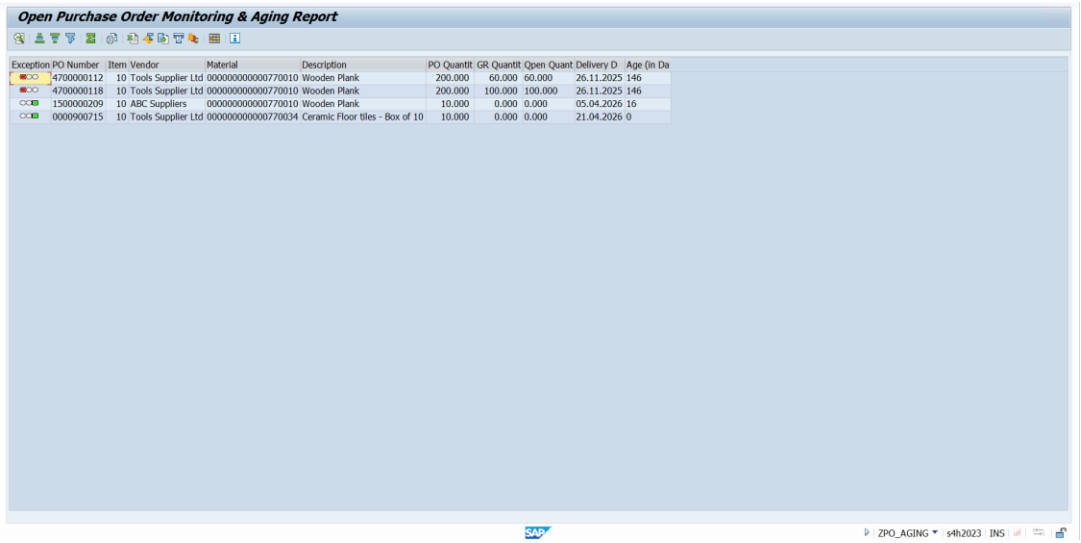
Technical Artifacts

- **Program Name:** ZOPEN_PO_AGING
- **Transaction Code:** **ZPO_AGING**
- **Tables Used:** EKKO · EKPO · EKET · LFA1
- **Output Type:** ALV Grid (REUSE_ALV_GRID_DISPLAY)
- **Created via:** SE38 + SE93
- **Authorization:** Standard user access in training system

Test Scenarios

- Execute report for Company Code and Plant with no PO number restriction
- Verify open PO lines appear with correct open quantity (PO qty minus GR qty)
- Confirm fully received lines are excluded from output
- Validate aging days calculated correctly against system date
- Check traffic light colors reflect correct aging thresholds (Green ≤30, Yellow ≤90, Red 90+)
- Export ALV output to Excel and verify all fields populate correctly
- Execute via custom T-code: ZPO_AGING and confirm report launches directly





Note: *The data in the screenshot is shown for a sample size of purchase orders (for demo purposes).*

ABAP Code Walkthrough — ZOPEN_PO_AGING

The report is structured into logical sections following standard ABAP conventions. Key sections are documented below. Full source code (268 lines) available on request — program is active and executable in SAP S/4HANA 2023 Training System.

1. Report Declaration & Table References

Declares the program, activates SLIS type pools for ALV, and references the database tables used in the SELECT statement.

```
REPORT ZOPEN_PO_AGING.

TYPE-POOLS: slis.

TABLES: ekko, ekpo, eket, lfa1.
```

2. Selection Screen

Allows the user to filter by Company Code (BUKRS), Plant (WERKS), and optionally by PO Number range. Keeps the selection screen simple and user-friendly.

```
PARAMETERS: p_bukrs TYPE ekko-bukrs,

            p_werks TYPE ekpo-werks.

SELECT-OPTIONS: s_ebeln FOR ekko-ebeln.
```

3. Internal Structure & Data Declaration

Defines the internal structure `ty_po` to hold all output fields including calculated fields (open quantity, aging days, traffic light status). `gt_po` is the output table; `gs_po` is the work area.

```
TYPES: BEGIN OF ty_po,
         ebeln  TYPE ekko-ebeln,       "PO Number
         ebelp  TYPE ekpo-ebelp,       "Item
         lifnr  TYPE ekko-lifnr,       "Vendor
         name1  TYPE lfa1-name1,       "Vendor Name
         matnr  TYPE ekpo-matnr,       "Material
         txz01  TYPE ekpo-txz01,       "Description
         menge  TYPE ekpo-menge,       "PO Quantity
         wemnq  TYPE eket-wemng,       "GR Quantity
         openq  TYPE ekpo-menge,       "Open Quantity
         eindt  TYPE eket-eindt,       "Delivery Date
         age    TYPE char10,           "Aging Days
         light  TYPE c LENGTH 1,       "Traffic Light
       END OF ty_po.

DATA: gt_po   TYPE TABLE OF ty_po,
      gs_po   TYPE ty_po.
```

4. Data Fetch — Open PO Lines

Joins EKKO, EKPO, LFA1, and EKET in a single SELECT to retrieve all open PO lines matching the selection criteria. Filters out deleted lines (`loekz = ''`) and already fully-received lines (`elikz = ''`).

```
SELECT a~ebeln b~ebelp a~lifnr c~name1
       b~matnr b~txz01 b~menge d~wemng d~eindt
  INTO CORRESPONDING FIELDS OF TABLE gt_po
  FROM ekko AS a
  INNER JOIN ekpo AS b ON a~ebeln = b~ebeln
  INNER JOIN lfa1 AS c ON a~lifnr = c~lifnr
  INNER JOIN eket AS d ON b~ebeln = d~ebeln
                       AND b~ebelp = d~ebelp
  WHERE a~bukrs   = p_bukrs
    AND b~werks   = p_werks
    AND a~ebeln  IN s_ebeln
    AND b~loekz   = ''
    AND b~elikz   = ''.
```

5. Open Quantity Calculation & Aging Logic

Loops through the fetched data to calculate the open quantity (PO qty minus GR qty) and aging in days using SY-DATUM. Lines fully received (open qty ≤ 0) are deleted from the output table.

```
DATA lv_age TYPE i.

LOOP AT gt_po INTO gs_po.

  "Open Quantity
  gs_po-openq = gs_po-menge - gs_po-wemnq.
  IF gs_po-openq < 0.
    gs_po-openq = 0.
  ENDIF.

  "Age Calculation
  lv_age    = sy-datum - gs_po-eindt.
  gs_po-age = lv_age.

  MODIFY gt_po FROM gs_po INDEX sy-tabix
               TRANSPORTING openq age light.

ENDLOOP.

DELETE gt_po WHERE openq <= 0.
```

6. Traffic Light Logic

Assigns a traffic light indicator to each PO line based on aging days — Green (≤30 days), Yellow (31–90 days), Red (90+ days). This drives the visual status column in the ALV output.

```
"Traffic Light Logic
  IF lv_age <= 30.
    gs_po-light = '3'.    "Green
  ELSEIF lv_age <= 90.
    gs_po-light = '2'.    "Yellow
  ELSE.
    gs_po-light = '1'.    "Red
  ENDIF.
```

7. ALV Field Catalog & Layout

Builds the ALV field catalog dynamically using a reusable `add_field` FORM routine. Layout settings enable zebra striping, column width optimisation, and the traffic light status column.

```
PERFORM build_fieldcat.

gs_layout-zebra             = 'X'.
gs_layout-colwidth_optimize = 'X'.
gs_layout-window_titlebar   = text-001.
gs_layout-lights_fieldname  = 'LIGHT'.
```

Field catalog build:

```
FORM build_fieldcat.
  PERFORM add_field USING 'LIGHT' 'Status'.
  PERFORM add_field USING 'EBELN' 'PO Number'.
  PERFORM add_field USING 'EBELP' 'Item'.
  PERFORM add_field USING 'NAME1' 'Vendor'.
  PERFORM add_field USING 'MATNR' 'Material'.
  PERFORM add_field USING 'TXZ01' 'Description'.
  PERFORM add_field USING 'MENGE' 'PO Quantity'.
  PERFORM add_field USING 'WEMNG' 'GR Quantity'.
  PERFORM add_field USING 'OPENQ' 'Open Quantity'.
  PERFORM add_field USING 'EINDT' 'Delivery Date'.
  PERFORM add_field USING 'AGE'   'Age (in Days)'.
ENDFORM.
```

8. ALV Grid Display

Calls REUSE_ALV_GRID_DISPLAY to render the output with the configured layout and field catalog. Passes `gt_po` as the output table.

```
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
  EXPORTING
    I_CALLBACK_PROGRAM      = 'sy-repid'
    I_CALLBACK_USER_COMMAND = 'USER_COMMAND'
    IS_LAYOUT               = gs_layout
    IT_FIELDCAT             = gt_fieldcat
  TABLES
    t_outtab                = gt_po.
```